

Abstract No.

Single-cell RNA-sequencing in tumour-bearing humanized mice recapitulates human meningioma and refines preclinical drug discovery.

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NF2-related schwannomatosis frequently leads to the development of multiple meningiomas responsible for increased morbidity and mortality. Until now, patient-derived xenograft (PDX) models have been generated in immunocompromised mice, which lack a robust immune-cell tumour microenvironment, a critical driver of meningioma tumorigenesis and progression. For the first time, we are using humanized mice engrafted with CD34+ hematopoietic stem cells prior to meningioma PDX, to analyse the tumour immune cell compartment by single-cell RNA-sequencing (scRNA-seq), to inform how concordant immune cell infiltration is to human meningiomas and also to identify immunotherapeutic targets and generate a refined model for NF2-deficient meningiomas.

Sc-RNAseq revealed accurate delineation of mouse, human meningioma and human immune cells. KT21-MG1, NCH93 and CH157 meningioma PDXs grew with significantly differential growth kinetics and tumour burdens compared to PDXs in non-humanized controls. ScRNA-seq revealed abundances and expression profiles of PDX immune infiltrates: for example, in KT21-MG1 PDXs (n=5), human immune cells accounted for on average 42% of total microenvironment cells with macrophages being the most abundant, followed by dendritic cells, NK cells and T cells. M2-polarized macrophages in the tumour expressed high levels of SPP1, which positively regulates macrophage infiltration and immune checkpoint activity. T cells and macrophages expressed immune checkpoint markers PD1, PD-L1 and CTLA4. We are currently benchmarking brigatinib in humanized and naïve controls and will compare to immunotherapies in vivo.

Humanized mice meningioma PDXs were viable with no evidence of graft-versus-host disease. ScRNA-seq revealed immune-cell infiltrates concordant with human NF2-null meningioma. Interestingly, we saw that top-upregulated antigens in a human immune cell atlas were also expressed abundantly in humanized PDXs. Overall, humanized meningioma PDX mice represent a major refinement to existing immunocompromised models, and incorporating them into NF2-null drug discovery pipelines has the potential to yield more powerful and translational drug efficacy data.



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